

Product Brief: AI Twi Audio Transcriber

1. Project Overview

Project Idea

The AI Twi Audio Transcriber is an AI-powered accessibility platform designed to convert spoken Asante Twi audio recordings into understandable English text. The platform is built to help individuals who do not speak or fully understand Twi access, comprehend, and utilize information contained within audio recordings.

The platform processes general audio recordings such as interviews, meetings, training sessions, public speeches, and community discussions, transforming them into actionable English content.

The AI Twi Audio Transcriber serves as a communication bridge between Twi speakers and non-speakers, reducing language barriers and enabling greater participation within Ghana's multilingual environment.

Mission

To transform spoken Asante Twi into accessible and understandable English text, enabling effective communication, collaboration, and information sharing between Twi speakers and non-speakers where language barriers frequently disrupt workflows and day-to-day operations.

Vision

To build the premiere, highly localized speech-to-text platform for West African languages, originating with a hyper-focused linguistic parsing engine for Asante Twi.

2. The Problem

Ghana is home to a diverse linguistic population, with Twi being one of the most widely spoken languages. Despite this, many professionals, researchers, expatriates, and residents are not fluent in Twi and frequently encounter situations where important information is communicated through spoken Twi recordings.

Common examples include:

- Workplace meetings
- Recorded instructions
- Interviews
- Community engagements
- Training sessions
- Podcasts and broadcasts

For non-Twi speakers, these recordings often present significant challenges:

- Difficulty understanding spoken content
- Dependence on translators or colleagues for clarification
- Time spent replaying recordings repeatedly
- Increased risk of miscommunication
- Limited accessibility to important information

While existing speech recognition and translation technologies exist, most are designed for enterprise infrastructure, developer integrations, legal archiving, or formal transcription workflows. Few solutions specifically address the everyday communication challenges faced by non-Twi speakers.

3. Solution

The AI Twi Audio Transcriber was conceived to address a practical communication challenge experienced by many individuals living and working in Ghana.

The core objective is not simply to convert speech into text. The objective is to ensure that users who do not understand Twi can still access, understand, and act upon information contained within spoken recordings.

The platform addresses this problem through several key principles:

- **Translation-First Accessibility:** Many existing solutions require users to first view a Twi transcript before translating it into English. For individuals who cannot read or understand Twi, this additional step provides little value. The AI Twi Audio Transcriber prioritizes English comprehension by making understandable English output the primary user-facing result.
 - **Reduced Communication Barriers:** The platform eliminates the need for users to depend on colleagues, friends, or professional translators to understand recorded content.
 - **Improved Productivity:** Users can quickly extract information from recordings without repeatedly replaying audio files or manually documenting key points.
 - **Greater Inclusivity:** Organizations, NGOs, businesses, and other institutions can engage more effectively with multilingual audiences by ensuring spoken content is accessible to all stakeholders.
 - **Real-World Language Understanding:** The platform is designed to accommodate informal speech patterns, conversational Twi, and code-switching between Twi and English that commonly occur in everyday communication.
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4. Target Audience

Primary Users

- Non-Twi-speaking professionals
- Expatriates living in Ghana
- Researchers and field workers
- Individuals working in multilingual environments

Secondary Users

- NGOs and development organizations
 - Government agencies
 - Journalists and media professionals
 - Corporate organizations
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5. Core Features (MVP)

The initial release of the AI Twi Audio Transcriber will focus on delivering a streamlined and highly accessible user experience.

- **Audio Upload:** Users can upload audio recordings in supported formats such as MP3, WAV, and OGG.
- **Audio Playback:** Users can listen to uploaded recordings directly within the platform.
- **AI-Powered Twi Recognition:** The platform automatically processes spoken Twi audio and identifies spoken content.
- **Twi-to-English Translation:** The system converts recognized Twi speech into understandable English text.
- **English-First Output:** Users receive clear English text optimized for readability and comprehension.
- **Copy and Export:** Users can copy translated text or export results for future reference.
- **Error Handling and Feedback:** The system provides clear feedback when audio quality, file format, or processing issues affect results.